

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1104

Louisiana Pacific 138 kV
July 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1104. The Interconnection Customer proposes a 76.0 MW solar photovoltaic facility interconnecting at the Louisiana Pacific 138 kV substation in the West Virginia area. The request was received on November 14, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 336 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,496,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Oasis Substation – Millbrae 138.0 kV	Halsey – Border Steel 138.0 kV	112.2	198	37.1
Huston – Goosport 138.0 kV	Cedar Crest Substation – Louisiana Pacific 138.0 kV Line	107.3	224	72.9
Tres Vias – Novato 138.0 kV	Novato – Rio Grande City 138.0 kV	104.1	245	8.8
Wareham Vly – Oasis 138.0 kV	Halsey – Border Steel 138.0 kV	100.3	224	37.1
Border Steel – Wareham Valley 138.0 kV	Border Steel – Halsey 138.0 kV Line	100.0	228	37.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Oasis Substation – Millbrae 138.0 kV	Rebuild line with higher-capacity conductor	\$5,981,000	38% (\$2,287,000)
Huston Substation – Goosport 138.0 kV	Reconductor line	\$5,403,000	28% (\$1,535,000)
Tres Vias – Novato 138.0 kV	Reconductor line	\$1,170,000	16% (\$187,000)
Wareham Vly – Oasis 138.0 kV	Reconductor line	\$1,755,000	38% (\$671,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Border Steel – Wareham Valley 138.0 kV	Reconductor line	\$2,134,000	38% (\$816,000)
	Total allocated to project		\$5,496,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.